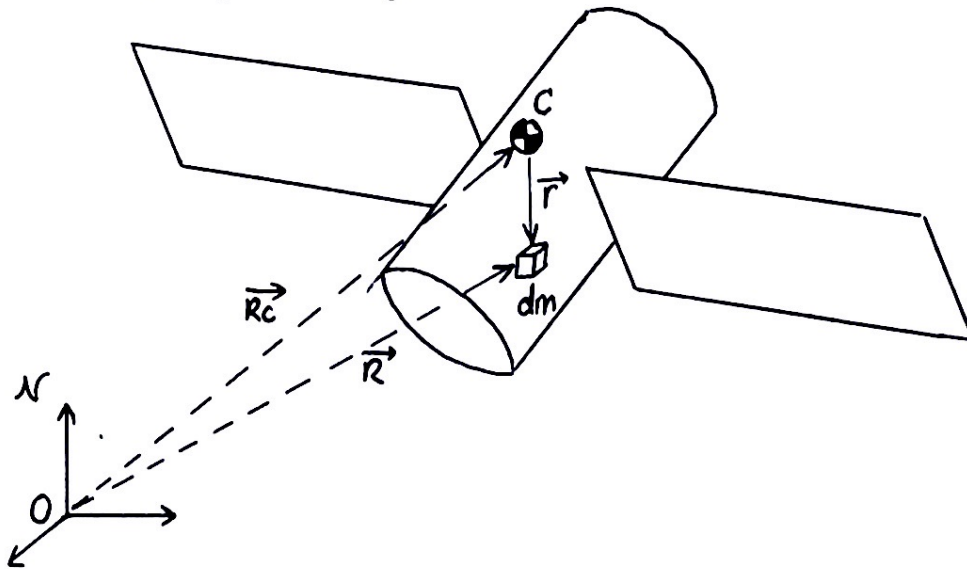


KINETICS: STUDYING SPACECRAFT MOTION

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Consider the rigid spacecraft system B shown:



A) Express the angular momentum vector $\vec{H}_C$ about the spacecraft center of mass as an integral over the body B

The angular momentum $\vec{H}_C$ of the rigid body B about its center of mass C is defined by the following integral.

$$\vec{H}_C = \int_B \vec{r} \times \dot{\vec{r}} \, dm \rightarrow \vec{H}_C = \int_B \vec{r} \times \dot{\vec{r}} \, dm$$

Since the spacecraft is rigid, the relative velocity is given by the kinematic relationship:

$$\dot{\vec{r}} = \vec{\omega} \times \vec{r} \rightarrow \text{Substituting: } \vec{H}_C = \int_B \vec{r} \times (\vec{\omega} \times \vec{r}) \, dm$$

B) Derive the inertial derivative of the $\vec{H}_C$ integral expression. Prove that this equals to the torque acting about the spacecraft center of mass. Make use of the fact up front that this spacecraft is rigid.

We start from the integral expression for the angular momentum about the center of mass C derived in Part A:

$$\vec{H}_C = \int_B \vec{r} \times \dot{\vec{r}} \, dm$$

We calculate the inertial time derivative by applying the chain rule inside the integral.

$$\dot{\vec{H}}_C = \frac{d}{dt} \int_B (\vec{r} \times \dot{\vec{r}}) \, dm = \int_B (\dot{\vec{r}} \times \dot{\vec{r}}) \, dm + \int_B (\vec{r} \times \ddot{\vec{r}}) \, dm = \int_B (\vec{r} \times \ddot{\vec{r}}) \, dm$$

This term equals to zero

From geometry, we know the inertial position is: $\vec{R} = \vec{R}_C + \vec{r} \rightarrow \ddot{\vec{r}} = \ddot{\vec{R}} - \ddot{\vec{R}}_C$
Substituting into the integral. $\dot{\vec{H}}_C = \int_B \vec{r} \times (\ddot{\vec{R}} - \ddot{\vec{R}}_C) \, dm$

$$\frac{d^2}{dt^2} \vec{r} = \ddot{\vec{R}}|_N - \ddot{\vec{R}}_C|_N$$

$$\dot{\vec{H}}_C = \int_B (\vec{r} \times \ddot{\vec{R}}) \, dm - \left(\int_B \vec{r} \, dm \right) \times \ddot{\vec{R}}_C = \int_B (\vec{r} \times \ddot{\vec{R}}) \, dm$$

By the definition of the center of mass, the first moment of mass relative to the center of mass itself is zero

Using Newton's second law: $d\vec{F} = \ddot{\vec{R}} dm$, the expression reduces to:

$$\dot{\vec{H}}_c = \int_B (\vec{r} \times \ddot{\vec{R}}) dm = \int_B \vec{r} \times d\vec{F} = \vec{\tau}_c$$

Since the definition of net torque about the center of mass is $\vec{\tau}_c = \int_B \vec{r} \times d\vec{F}$, it is proven that: $\dot{\vec{H}}_c = \vec{\tau}_c$

c) Starting with the general kinetic energy expression $T = \frac{1}{2} \int_B (\dot{\vec{R}} \cdot \dot{\vec{R}}) dm$, derive the kinetic energy expression that expresses in terms of the energy of the center of mass and the rotational energy about the center of mass. Make use of the fact up front that this spacecraft is rigid.

(Hint: recall the vector identity $(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$)

From the system's geometry, the inertial position is: $\vec{R} = \vec{R}_c + \vec{r}$

Taking the time derivative: $\dot{\vec{R}} = \dot{\vec{R}}_c + \dot{\vec{r}}$

Substituting into the general kinetic energy expression: $T = \frac{1}{2} \int_B (\dot{\vec{R}}_c + \dot{\vec{r}}) \cdot (\dot{\vec{R}}_c + \dot{\vec{r}}) dm$

Expanding the dot-products:

$$T = \underbrace{\frac{1}{2} \int_B (\dot{\vec{R}}_c \cdot \dot{\vec{R}}_c) dm}_I + \underbrace{\int_B (\dot{\vec{R}}_c \cdot \dot{\vec{r}}) dm}_II + \underbrace{\frac{1}{2} \int_B (\dot{\vec{r}} \cdot \dot{\vec{r}}) dm}_III$$

Analyzing each term:

I) $\dot{\vec{R}}_c$ is constant over the mass, so it moves outside the integral:

$$\frac{1}{2} \int_B (\dot{\vec{R}}_c \cdot \dot{\vec{R}}_c) dm = \frac{1}{2} \dot{\vec{R}}_c \cdot \dot{\vec{R}}_c \int_B dm = \frac{1}{2} M \cdot \dot{\vec{R}}_c \cdot \dot{\vec{R}}_c$$

II) $\int_B (\dot{\vec{R}}_c \cdot \dot{\vec{r}}) dm = \dot{\vec{R}}_c \cdot \frac{d}{dt} (\int_B \vec{r} dm) = 0$

III) Rotational energy about the center of mass: $T_{rot} = \frac{1}{2} \int_B (\dot{\vec{r}} \cdot \dot{\vec{r}}) dm$

Since the spacecraft is rigid, we can use: $\dot{\vec{r}} = \vec{\omega} \times \vec{r}$ Substituting:

$$T_{rot} = \frac{1}{2} \int_B (\vec{\omega} \times \vec{r}) \cdot (\vec{\omega} \times \vec{r}) dm \leftarrow \text{We can apply } (\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$(\vec{\omega} \times \vec{r}) \cdot \dot{\vec{r}} = \vec{\omega} \cdot (\vec{r} \times \dot{\vec{r}}) \rightarrow T_{rot} = \frac{1}{2} \vec{\omega} \cdot \underbrace{\int_B (\vec{r} \times \dot{\vec{r}}) dm}_{\vec{H}_c} = \frac{1}{2} \vec{\omega} \cdot \vec{H}_c$$

Finally, we obtain:

$$T = \frac{1}{2} M \dot{\vec{R}}_c \cdot \dot{\vec{R}}_c + \frac{1}{2} \vec{\omega} \cdot \vec{H}_c$$

NOTE: $\dot{\vec{R}}_c$ is the velocity of the center of mass, which is uniform for all mass elements dm of the rigid body, making it independent of the integration over the body.

Since B is a rigid body, $\vec{\omega}$ is a property of the entire frame and is identical for every dm . Thus, it can be created as a constant relative to the body integral.